



CS957 Research Project Proposal

University of Strathclyde — Department of Computer and Information Sciences

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1 Introduction and Literature Review

1.1 Background

The UK has been affected by climate change in ways such as flooding, which is one of the most disruptive effects. The occurrence and intensity of flooding activities have grown tremendously in the past decades due to the changes in the nature of precipitations and the rise in sea level [1]. Glasgow and the west of Scotland in general, is especially vulnerable: the River Clyde catchment is subject to frequent high-flow events, and the densification of the urban areas has decreased the natural drainage capacity of the city.

Even though there are comprehensive environmental data available, such as Scottish Environment Protection Agency (SEPA) flood boundary shapefiles, Met Office HadUK-Grid climate data, and real-time air quality data available on OpenAQ, there has always been a disconnect between the availability of such data and the ability of people to access it. Citizens, local civil societies and those in the various decision-making units find it difficult to translate the technical risk mapping into individual or group actions [2], [3].

The gap in this project is filled by creating an interactive machine learning dashboard, which uses public data, trains a flood risk classifier, and presents the results with accessible advice to the non-specialist user. The study adheres to a Design Science Research (DSR) approach [4], in which the dashboard is a designed artefact, the value of which is measured on the basis of a criterion of usability by a non-expert audience.

1.2 Literature Review

The systematic search was performed in the academic literature of IEEE Xplore, Google Scholar, and Web of Science. Keywords were the combinations of flood risk prediction machine learning, geospatial decision support, climate risk communication, random forest hydrology, and public understanding environmental data.

1.2.1 Flood Risk Prediction by machine learning.

- **Ensemble classifiers to flood mapping:** Ensemble approaches, especially the random forests and gradient-boosted trees, have consistently outperformed better than single classifiers, in flood susceptibility mapping. Tehrany et al. showed that ensemble models can recall minority flood-event classes much more effectively than logistic regression or SVM baselines [5], and hence can be considered the right place to start this project.
- **Class imbalance of environmental data:** Cases of flood are infrequent compared to non-flood scenarios and result in systematic class imbalance. Traditional measures of accuracy do not apply here, the literature suggests using F1-score and Area Under the ROC Curve (AUC-ROC) as the main measures of evaluation of a flood classifier [5].
- **Model interpretability:** Interpretability is not only a usability issue there are direct trust and adoption implications. The standard procedure is now SHAP (SHapley Additive exPlanations) used to explain individual predictions of tree-based models [14], [15]. SHAP values, in this project, will be directly related to the goal of public understanding, by displaying the environment variables behind each risk prediction.

1.2.2 Decision Support Systems of Environmental Risk.

- **Prescriptive/descriptive DSS** The literature relating to DSS has differentiated between systems that suggest action (what should I do?) and systems that describe a situation (risk maps, flood boundaries). According to Rizzoli and Young, the main challenge of environmental decision support is the shift to the descriptive to the prescriptive one [2]. The majority of existing tools facing the public are still on the descriptive level.
- **Design science as evaluation framework** Planning Design science artefacts should not just be assessed based on the technical correctness but the usefulness and suitability to the problem environment [4]. The evaluation strategy in this project is an inherent result

of this framing, the dashboard is not finished until it is evaluated based on usability rules by appropriate users.

- **Disparity between climate adaptation tools:** Biesbroek et al. report that even high-budgeted climate adaptation programs have often failed to deliver technical outputs to behavioural change due to a lack of a communication layer designed to better reach non-specialist audiences [6]. In this project, that layer of communication is considered to be the technical input in terms of contribution.

1.2.3 Urban Flood Risk Geospatial Analysis.

- **Multi-source spatial data integration:** Urban flood risk analysis needs to integrate heterogeneous spatial data on various scales at varying coordinates. SEPA flood boundary information is in the form of vectors whereas HadUK-Grid is rasterised climate information at a spacing of 1km. Tomaszewski sets technical and cognitive requirements of the spatial data integration to non-professional decision support [7].
- **Public engagement through interactive cartography:** Singleton and Spielman offer the evidence that compared with the static maps, the interactive web maps could help to make the people understand the spatial risk information better [8].

1.2.4 Risk Communication Recommendation Systems.

- **Rule-based vs. ML-based recommendation:** Ricci et al. distinguish between content-based and collaborative filtering approaches to recommendation and rule-based [9]. In the current project, the rule based approach will be best: the mapping between the level of flood risk and the action recommended is domain controlled (emergency guidance by SEPA and UK government) than the one learned through historical user preferences.
- **Gap in the literature:** The proposed research is expected to have a gap in the literature because there is no known system that brings together a trained ML flood risk classifier and a structured recommendation engine in a publicly accessible dashboard. According to Biesbroek et al., this prescriptive gap is specified directly by the fact that there are tools to predict or to communicate, but not both in a single interface [6].

1.2.5 Social Climate risk knowledge.

- **Cognitive barriers to probabilistic risk:** Spiegelhalter shows that probabilistic results (e.g. 35% flood probability or other such results) are mistakenly interpreted systematically by non-experts who will translate them into binary certainties [10]. The discovery itself makes a direct impact on the dashboard design: risk outputs are going to be displayed in the form of categorical levels (Low / Medium / High) and plain-language description instead of raw probabilities.
- **Engagement drivers (Trust and personal relevance):** According to Renn, risk communication is best done when it relates aggregate risk information to personally relevant situations and when conveyed by an institution with which one can be associated with (Trust) [11].
- **Usability heuristics of tools that are visible to the people:** Nielsen 10 usability heuristics [20] offer the analysis platform to the dashboard. Each of the 10 heuristics will be systematically used in the usability evaluation and a heuristic evaluation score sheet will be created as a documented output in addition with the participant walkthrough notes.

1.3 Research Gaps

The literature review reveals three gaps that are filled in this project:

- No integrated system exists in Glasgow that has a trained ML based flood risk classifier, structured recommendation engine, and an interactive public facing dashboard.

- The current decision support tools on flood risk are on the level of description and do not transform the risk outputs into prescribed actions that can be seen by non-specialist users [2], [6].
- No reported research has used SHAP feature importance to relate the ML flood risk predictions with human-understandable interpretations in the context of an interactive dashboard system [14].

1.4 Research Contribution

The study will be relevant to climate informatics and decision support systems as it incorporates machine learning-based predictive of flood risk, SHAP-based interpretability, and an organized recommendation system into a single dashboard that can be accessed by the general public. In contrast to the current systems which still stay on the descriptive level i.e. showing risk information and not providing actionable information, this work provides a prescriptive, user-friendly artefact to enhance the general population knowledge and hence make practical decisions in relation to climate adaptation.

The scope will be narrowed down to batch data processing and one case study (Glasgow) city, such that the system is realistically attainable and assessable within the three month time frame at hand.

2 Research Aim, Objectives and Research Questions

2.1 Research Aims

1. To create a free, open-source environmental data-based end-to-end pipeline of flood risk prediction in the city or region of Glasgow by using machine learning classifiers that have been validated.
2. To create a rule based recommendation engine, which will then convert the risk level that is predicted into plain language and actionable advice to be used by non-specialist users of the product.
3. To evaluate the usability of the resulting Streamlit dashboard using evaluation framework and implement in a systematic manner.

2.2 Research Objectives

The objectives below are made in accordance with SMART principles (Specific, Measurable, Achievable, Relevant, Time-bound) and are aimed at being accomplished within the timeframe of the estimated hours of work on an MSc dissertation project.

- The goal is to create a good F1-score of a baseline classifier at the end of Week 6 by utilizing training and testing on the held-out test set.
- Use SHAP analysis to generate feature importance outputs of the top three important environmental predictors of flood risk by the conclusion of Week 6.
- Within the scope of Week 8, create and pilot a rule-based recommendation engine on Low, Medium and High-risk types.
- Develop a working Streamlit dashboard that combines risk map, prediction result and recommendation panel by the end of Week 10.

2.3 Research Questions

The following three research questions can be directly tested in the project timeline.

RQ1: How much can a machine learning classifier predict the risk of floods at the neighbourhood level in Glasgow, and what are the environmental factors that can best predict the risk of floods?

RQ2: Does the rule-based recommendation engine generate recommendations that are rated by evaluators as comprehensible, applicable and related by a non-specialist audience?

RQ3: Is the engine dashboard usable to a level that it passes the 10 heuristic evaluation framework.

3 Research Methods

The research is a combination of quantitative and qualitative research.

- This is necessary as the project is technically oriented (model building and evaluation) and human-oriented (usability and communication effectiveness) and each of these aspects demands different approaches to be assessed in a meaningful way.
- Figure below depicts the general structure of the system which illustrates the five-layered CARE (Climate Awareness and Risk Evaluation) pipeline in between raw data digestion and dashboard output.
- The architecture is also modular with deliberate design: All the layers can be changed or replaced without impacting the overall research goals. In the case of problems with data availability, fallback sources (Environment Agency, Open Meteo) can be used in place of primary sources.
- In case the main model is not performing adequately, the XGBoost or LightGBM can be replaced in the modelling layer without altering the upstream and downstream of the pipeline.

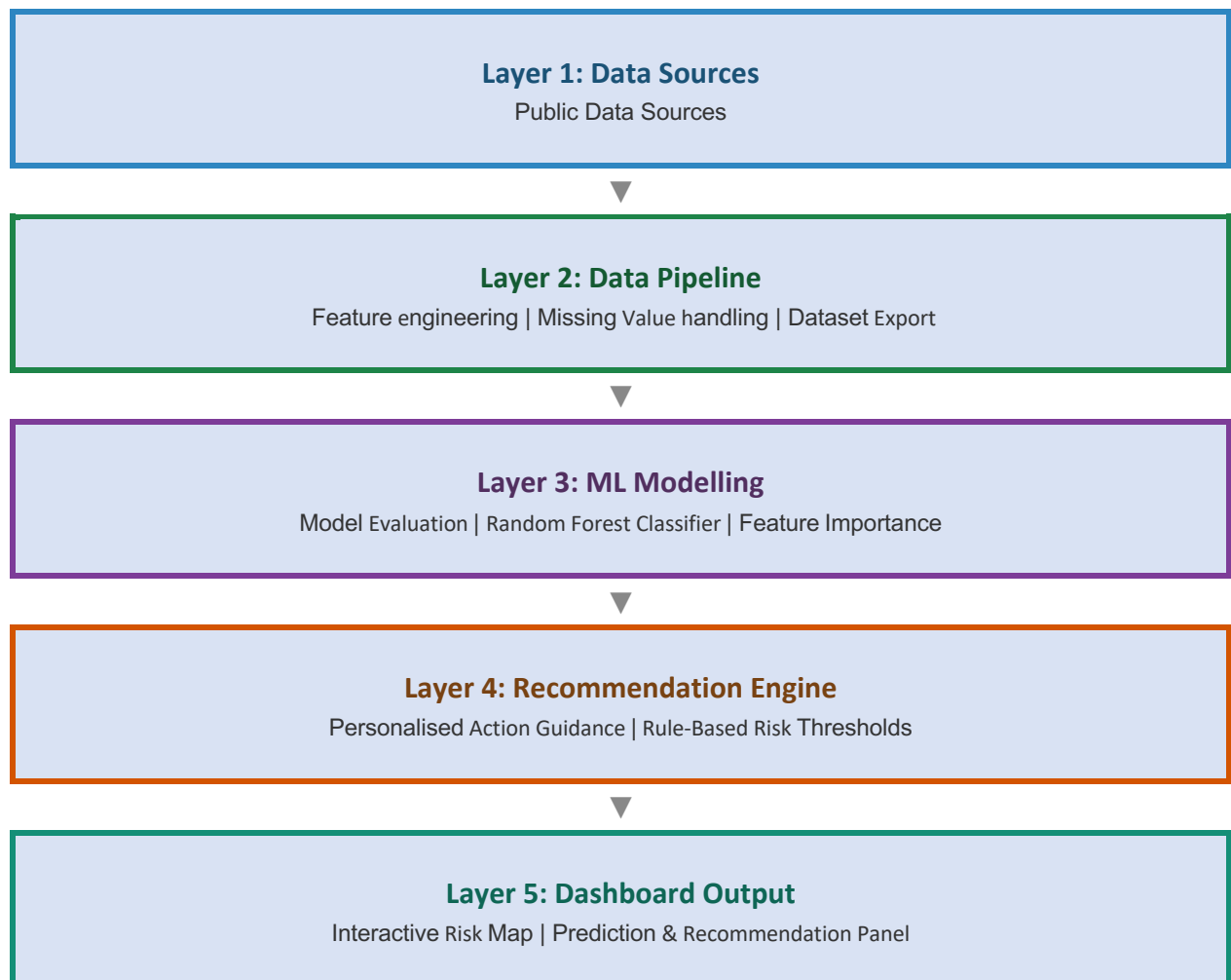


Figure 1: CARE System Architecture - Five-layer pipeline between the sources of data and the dashboard output.

3.1 Quantitative Analysis

3.1.1 Data Collection and Preprocessing data

The data will be collected using public data sources and then pre-processed by conducting the research.

- **Primary data sources:** Any Openly available data set like SEPA Flood Maps of Scotland, UK Met Office HadUK-Grid, OpenStreetMap, OpenAQ or Kaggle etc.
- **CRS alignment:** There will be no spatial join until all the datasets are reprojected with Geo-Pandas. It is a technical requirement to be able to do the area and distance calculations across data sources.
- **Spatial joins** will be conducted to label and provide flood risk and climate features to every geographic unit.
- **Flexibility of the data:** In case the access to SEPA or HadUK-Grid data is lagged after Week 2, the Environment Agency Flood Map for Planning (England), OpenMeteo historical weather API or Kaggle are other sources that fit similar variables.

3.1.2 Performance Analysis

The main model will be a ML classifier, as it has already been successfully applied in conducting flood susceptibility tasks and can process class imbalance and correlated features without requiring a significant amount of preprocessing [\[12\]](#).

F1-score (macro-averaged), AUC-ROC and a confusion matrix will be used to measure performance because these measures provide a more truthful view than accuracy when the occurrence of floods in the data is rare.

SHAP values will be produced on every output [\[14, 15\]](#) to get an idea of what is making the predictions. The dashboard will contain a summary plot that will allow the user to understand what environmental factors are the most important contributors to the risk score in their location - this is the key idea behind the public understanding goal of project.

3.1.3 Usability Evaluation

- **Report evaluation:** Systematic evaluation of the final dashboard will use Nielsen heuristics [\[20\]](#). A heuristic evaluation score sheet will be generated that will rank each heuristic based on a 0 4 severity scale. This is the main output of the usability.
- **Participants:** The university community will be considered a convenience sample of 3 to 5 participants with no background on the field of flood risk or data science. This sample size will be suitable to undertake exploratory heuristic assessment - Nielsen shows that 3-5 evaluators will determine about 75% of usability issues, however, the outcomes will not be statistically generalised.
- **Procedure:** Each participant will be asked to complete a guided walkthrough (e.g. Find the flood risk level at the postcode G1 1XQ and what should the task advise to be done?). One can use think-aloud protocol. The sessions can be taped (with permission) to be analysed later using a thematic approach.

3.2 Qualitative Analysis

3.2.1 Literature Review Breakdown

In Section 1.2, the literature review addresses five thematic areas, namely: machine learning to forecast flood risk [\[5\]](#), [\[12\]](#), [\[13\]](#), [\[14\]](#), [\[15\]](#); environmental decision support systems [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[6\]](#); geospatial analysis [\[8\]](#), [\[7\]](#); recommendation systems[\[9\]](#); and public understanding of

climate risk [\[10\]](#), [\[11\]](#), [\[20\]](#). The review was done according to systematic search principles having documented inclusion criteria (Section 1.2 intro). It gives the design conceptual basis of all the project design decisions.

3.2.2 User Walkthroughs

The thematic analysis will be used to analyse qualitative data gathered by usability sessions according to Braun and Clarke [\[18\]](#). Codes will be created inductively using transcripts of walkthroughs and in the process of think-aloud.

The research may collect expert opinion on the risk communication design using published reports by SEPA [\[19\]](#) and Spiegelhalter [\[10\]](#), and any other interviews or policy reports available on flood risk communication in the Scottish scene.

4 Research Plans & Deliverables

4.1 Research Timeline

The estimated amount of work on an MSc. Project, consist of an introduction of five phases to the project. Week 1-4 (approximately 120 hours) Weeks 5-6 (approximately 70 hours) Weeks 7-10 (approximately 120 hours) Weeks 11-12 (approximately 40 hours) Weeks 12 write-up (approximately 40 hours).

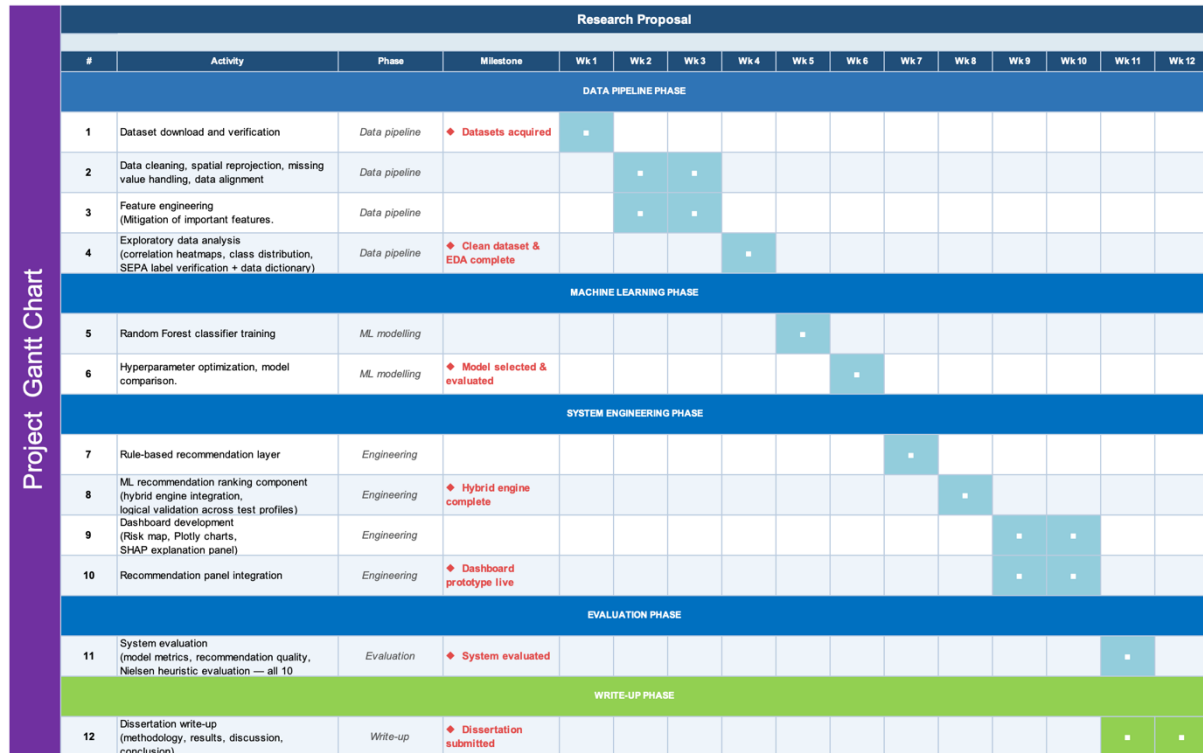


Figure 2: Gantt Chart of the project - 12-week colour-coded schedule with phase bands and milestones.

Week	Activity	Phase	Milestone
1	Data loading - download, check, all available datasets.	Data Pipeline	
2	Data cleaning: filters away the missing data and outliers, first EDA Data Pipeline.	Data Pipeline	
3	Spatial join - CRS alignment, feature generation Data Pipeline.	Data Pipeline	
4	Features engineering - encode targets, create train/test splits Data Processing	Data Pipeline	◆ Data Ready
5	Training and preliminary evaluation of the Baseline model training.	ML Modelling	
6	Optimization of Hyperparameter, SHAP analysis, model comparison.	ML Modelling	◆ Model Trained
7	Rule based suggestion system; Dashboard skeleton application	System Engineering	
8	Recommendation engine testing; start dissertation methodology chapter	System Engineering	◆ Rec. Engine Done

9	Prediction panel, dashboard, integration System engineering.	System Engineering	
10	Dashboard refinement, graphical design, end-to-end testing	System Engineering	◆ Dashboard Built
11	Heuristic assessment of 3 5 people; thematic analysis	Evaluation	◆ Evaluation Done
12	Final write-up, proof-read, preparation of submission Write-Up	Write-Up	◆ Submission

Table 1: Reflection of project schedule table

4.2 Risk and Contingency Plan

The seven most plausible risks to this project have been recorded in the table below based on the literature material on common failure modes in MSc dissertation and the technical issues particular to this methodology.

Risk Summary	Monitor	Action By	Likelihood	Impact	Priority	Mitigation
Data Access Delays	Difficulty accessing SEPA or HadUK-Grid data.	Week 1–2 check	Student	Medium	High	High
Geospatial Alignment Problems	CRS discrepancies between SEPA, OS, HadUK data lead to join failures Test joins	Test joins in Week 3	Student	Medium	High	High
Poor Model Performance	Random Forest F1 score not up to threshold because of the class imbalance	Assess at Week 6	Student	Low	High	Medium
Dashboard Timeline Squeeze	ML pipeline overruns with too little time to build Streamlit.	Weekly Gantt review	Student	Medium	Medium	Medium
Participant Recruitment for Evaluation	Challenge in recruiting 3-5 usability participants within the university community	Confirm by Week 10	Student	Low	Medium	Low
Write-Up Time Underestimated	Not enough time to write dissertation chapters	Begin drafting Week 8	Student	Low	Low	High

Table 2: Risk and Contingency plan table

4.3 Ethical Considerations

In this research, one will be in contact with open environmental data and usability testing on the human subjects.

The following requirements of ethics can be applied:

- The University of Strathclyde Departmental Ethics Committee (DEC) will be consulted in writing about the commencement of any of the participants by the CIS ethics portal. None of the activities of the participants will be conducted before the approval. *

- The participants will be provided with a Participant Information Sheet (PIS) that will refer to the purpose of the study, the right to be withdrawn without a penalty, and the purpose of using their data. At the start of the sessions, the informed consent will be signed.
- The used datasets are all open-access and do not involve any personal identifiable information. No real financial or individual details will be addressed.
- The dashboard has a social communications requirement in its risk communications. Spiegelhalter as well as Lundberg et al. indicate that probabilistic outputs should be presented in a manner that will not mislead [\[10\]](#) and [\[15\]](#) respectively. This will be put into calibration in regards to outputs that will be supported with explanatory text.

4.4 Deliverables

Besides the dissertation report, there will be six physical deliverables of the project:

- **Dissertation Report:** This is the primary report which summarizes the whole research process, findings as well as reflections. COTF. All other generations contain a cleaned Dataset and Feature Matrix: A documented data pipeline of preprocessing actions, in Geo-JSON and CSV, with the exception being the Automats.
- **Trained ML Model:** A serialised ML model that has been reported, evaluation (F1, AUC-ROC) and SHAP analysis results.
- **Recommendations Engine:** A table of guidelines in the form of a JSON, which holds the mapping of the risk levels with plain-language advice of what to do, the outcome of which is recorded.
- **Dashboard:** Interactive dashboard that functions (e.g.: Folium) with the risk map, prediction results, SHAP explanation and a recommendation panel, which can be locally deployed with a requirements.txt.
- **Heuristic Evaluation Report:** A scored sheet of Nielsen heuristics [\[20\]](#), participant walkthrough to notes, and the result of thematic analysis.

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